EVERY NOETHERIAN SEPARATED SCHEME OF DIMENSION AT MOST ONE CARRIES AN AMPLE INVERTIBLE SHEAF

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Ampleness can be checked on the reduction. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-ample-on-reduction (tag 0B3J). Carries ampleness of the lifted sheaf on the reduction back to X.
- Reduced Noetherian separated schemes of dimension at most one have an ample invertible sheaf. Stacks Project, Varieties, Lemma lemma-dim-1-noetherian-reduced-separated-has-ample (tag 09NY). Provides the starting ample sheaf on the reduction.
- Lifting invertible sheaves from a closed subscheme in dimension at most one. Stacks Project, Varieties, Lemma lemma-lift-line-bundle-from-reduction-dimension-1 (tag 0B3M). Produces the invertible module on X restricting to the ample sheaf on the reduction.

Source reference. Original lemma 09NX: sources/stacks-chow/tags.txt, source lines 12230–12230. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09NY: sources/stacks-chow/tags.txt, source lines 12231–12231. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09MS: sources/stacks-chow/tags.txt, source lines 12190–12190. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Proposition 0.1. *Let X be a Noetherian separated scheme of dimension 1. Then X has an ample invertible sheaf.*

Proof. Let $Z \subset X$ be the reduction of X. By Lemma 09NX the scheme Z has an ample invertible sheaf. Thus by Lemma 09NY there exists an invertible $\mathcal{O}_X$-module $\mathcal{L}$ on X whose restriction to Z is ample. Then $\mathcal{L}$ is ample by an application of Cohomology of Schemes, Lemma 09MS. □

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